

EX. NO: 1

Evidence Acquisition Using Access Data FTK Imager

Date: 09-08-2026

1. Aim

To acquire volatile memory and non-volatile storage evidence using Access Data FTK Imager and verify the integrity of the acquired disk image using cryptographic hash values.

2. Description

FTK Imager is a forensic acquisition and preview tool used to collect digital evidence. It can acquire volatile memory from a live Windows system and create forensic images of physical or logical storage devices.

In live acquisition, FTK Imager may be executed on the running evidence computer. For a powered-off or externally connected evidence drive, the source should be accessed through a write blocker whenever possible. A write blocker prevents accidental modification of the original evidence while allowing the investigator to read and acquire it.

This experiment contains two activities:

1. Volatile-memory acquisition from RAM.
2. Non-volatile physical-drive acquisition from an authorized USB device.

3. Software and Hardware Used

- Operating system: Windows
- Tool: AccessData FTK Imager 4.7.1.2
- Source device: JetFlash Transcend 8GB USB Device
- Source identifier: \\.\PHYSICALDRIVE1
- Image format: Raw (dd)

Part A: Volatile-Memory Acquisition

Step 1: Open the Memory Capture Window

FTK Imager was opened and the memory-capture function was selected.

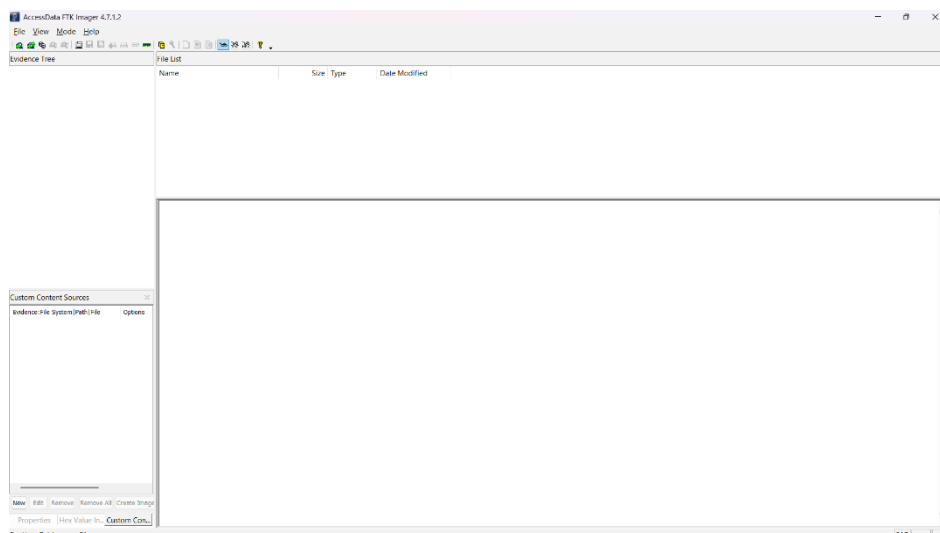


Figure 1: FTK Imager Memory Capture window.

Step 2: Configure Memory Capture

The following settings were used:

Setting	Value
Destination path	D:\n\df
Memory file	memdump.mem
Include pagefile	Enabled
Create AD1 file	Enabled
AD1 file	memcapture.ad1

The pagefile was included because it may contain memory pages written to disk by Windows. The AD1 option was enabled to create an FTK-compatible evidence container.

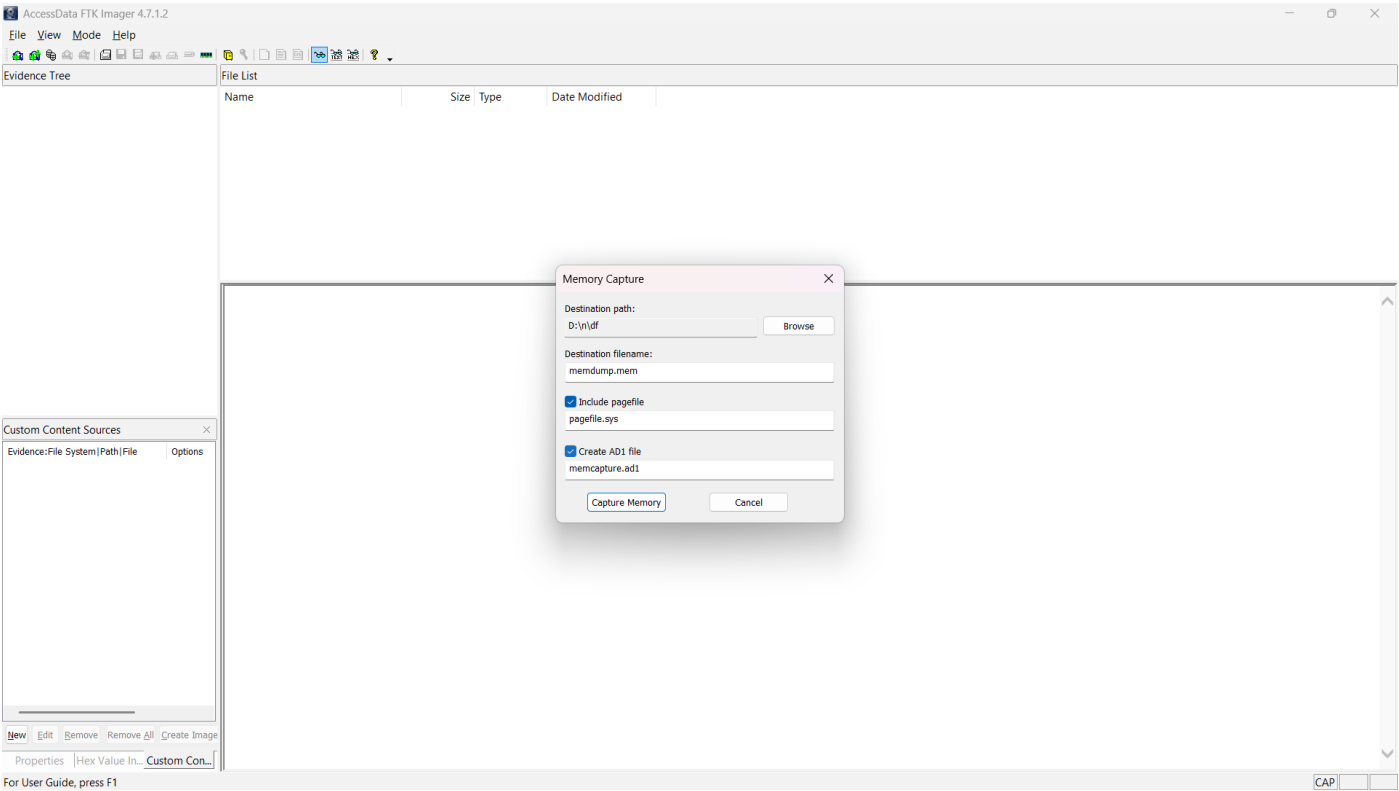
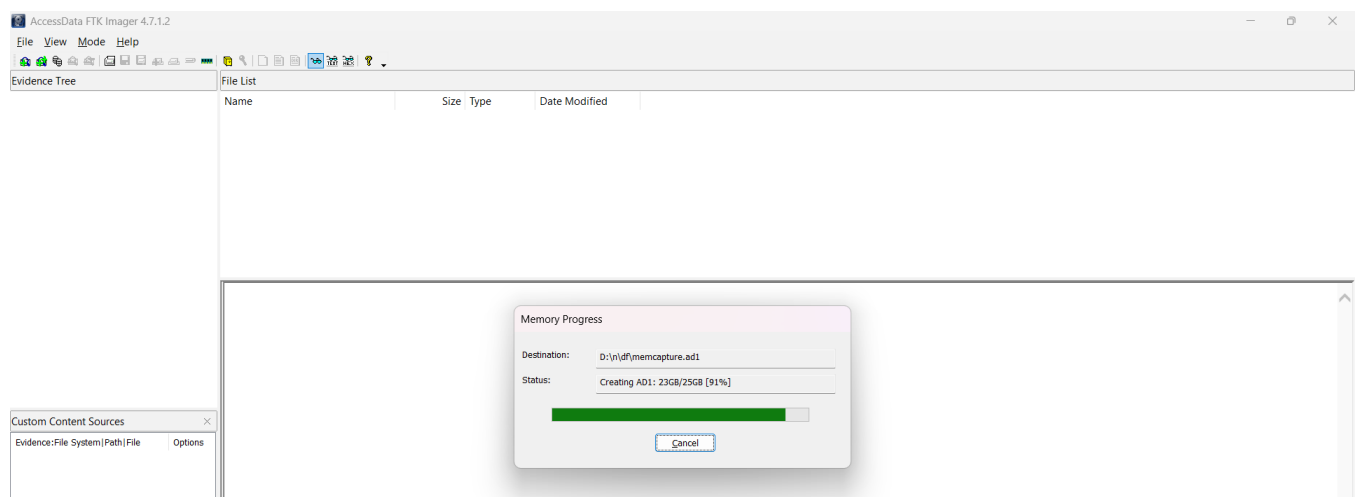
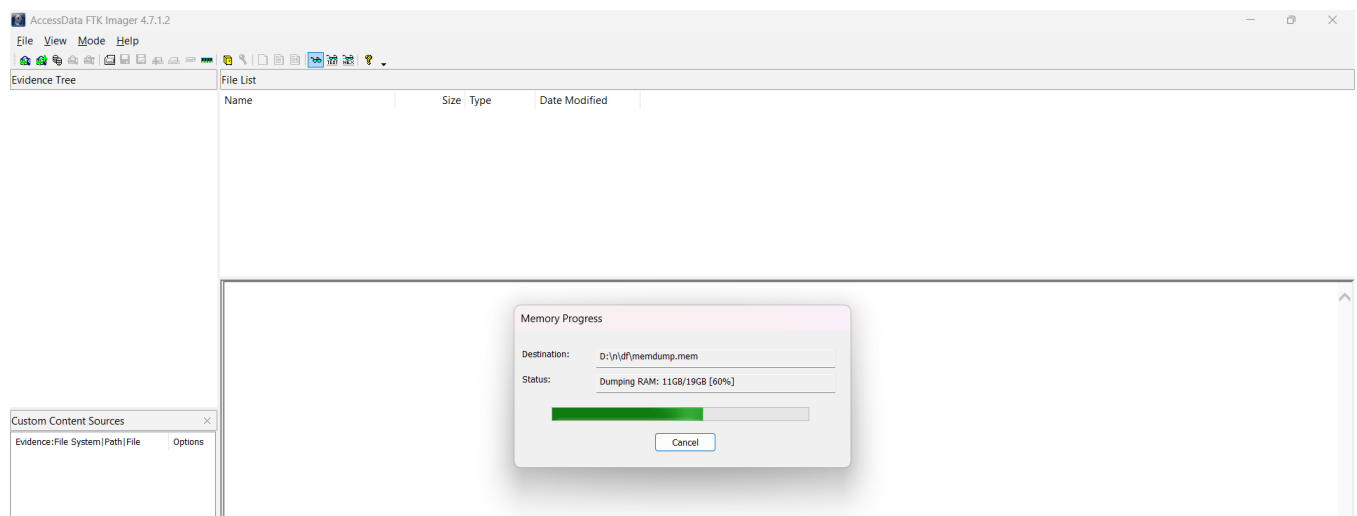


Figure 2: Memory Capture configuration showing the .mem, pagefile, and AD1 options.

Step 3: Capture RAM

The capture process was started. FTK Imager displayed the RAM-dumping progress, including the amount processed and the total memory size.



Step 4: Confirm Completion

FTK Imager reported:

text

Memory capture finished successfully

The memory output was written to:

text

D:\n\df\memdump.mem

The AD1 output was written as segmented files beginning with:

text

D:\n\df\memcapture.ad1

D:\n\df\memcapture.ad2

D:\n\df\memcapture.ad3

D:\n\df\memcapture.ad4

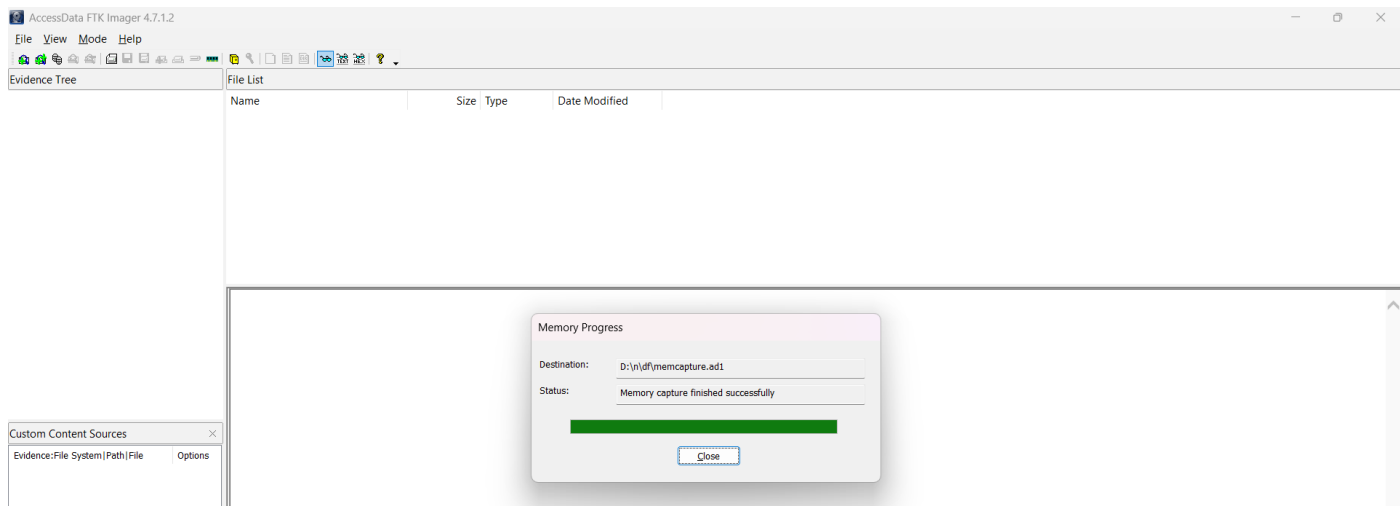


Figure 6: Successful completion of the memory capture.

Volatile-Memory Acquisition Report

The FTK report recorded the following checksums:

Hash	Value
MD5	e6718c721d0cfbe7ad41060c2b425ef7
SHA1	b403afb67550ce93c43c53a88dae2ed20dcd61f5

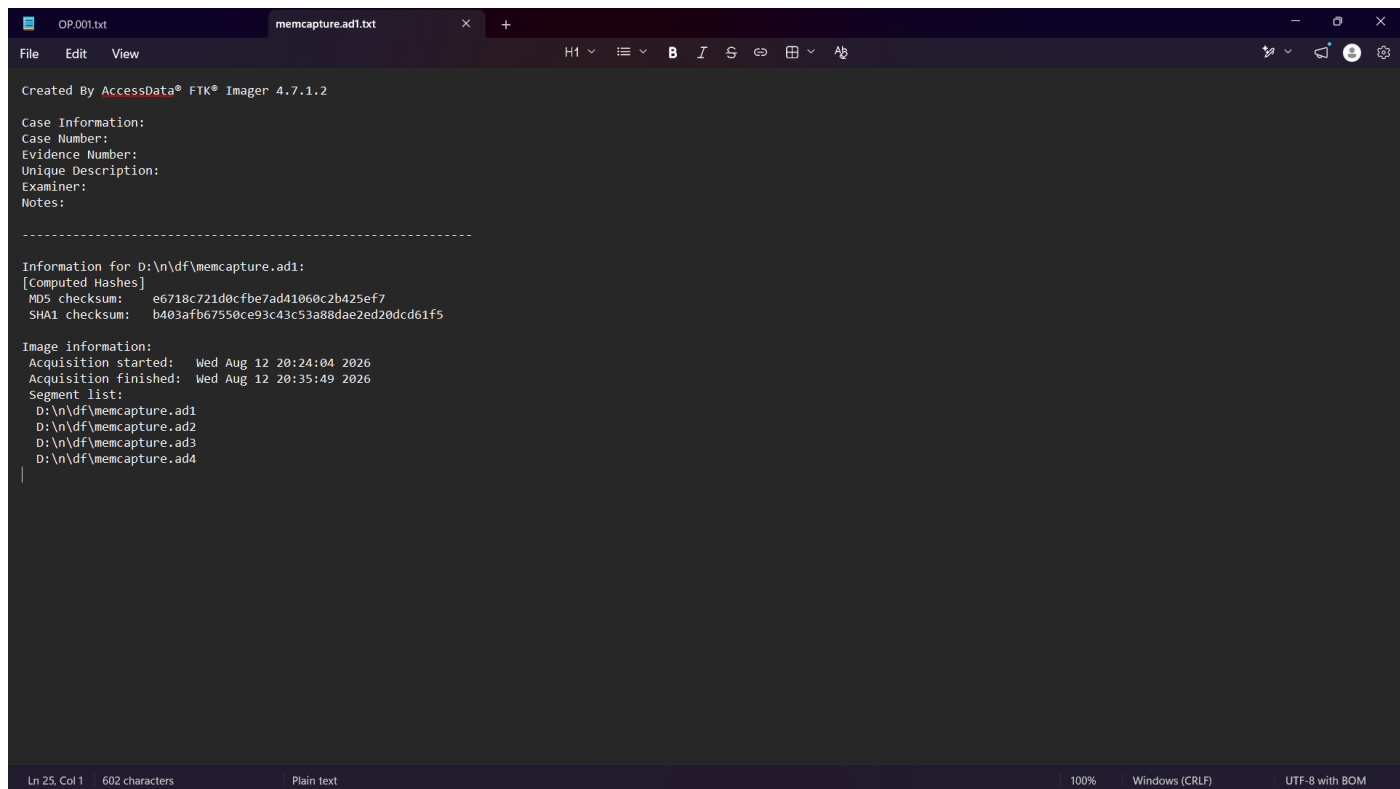


Figure 7: FTK memory-acquisition report showing the computed hashes and segment list.

Part B: Non-Volatile Physical-Drive Acquisition

Step 1: Select the Source Type

FTK Imager was opened and File → Create Disk Image was selected. In the Select Source window, Physical Drive was selected.

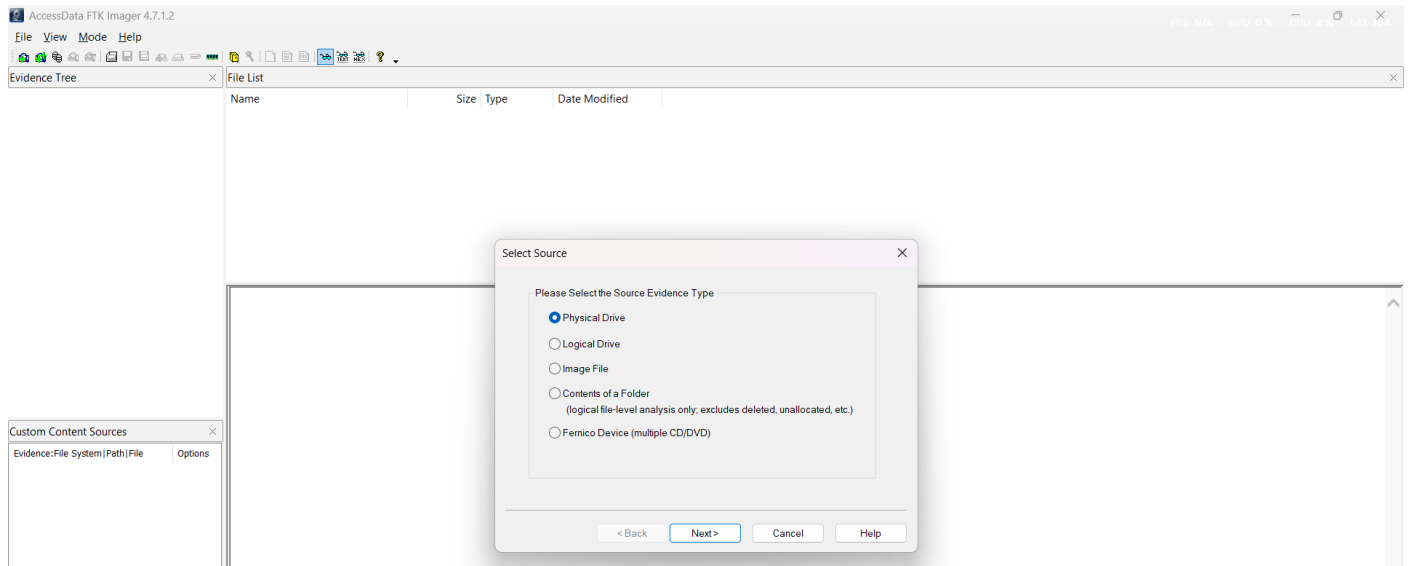


Figure 8: Selecting Physical Drive as the source evidence type.

Step 2: Select the Evidence Drive

The following drive was selected:

text

\\.\PHYSICALDRIVE1 - JetFlash Transcend 8GB USB Device

The internal Samsung drive was not selected.

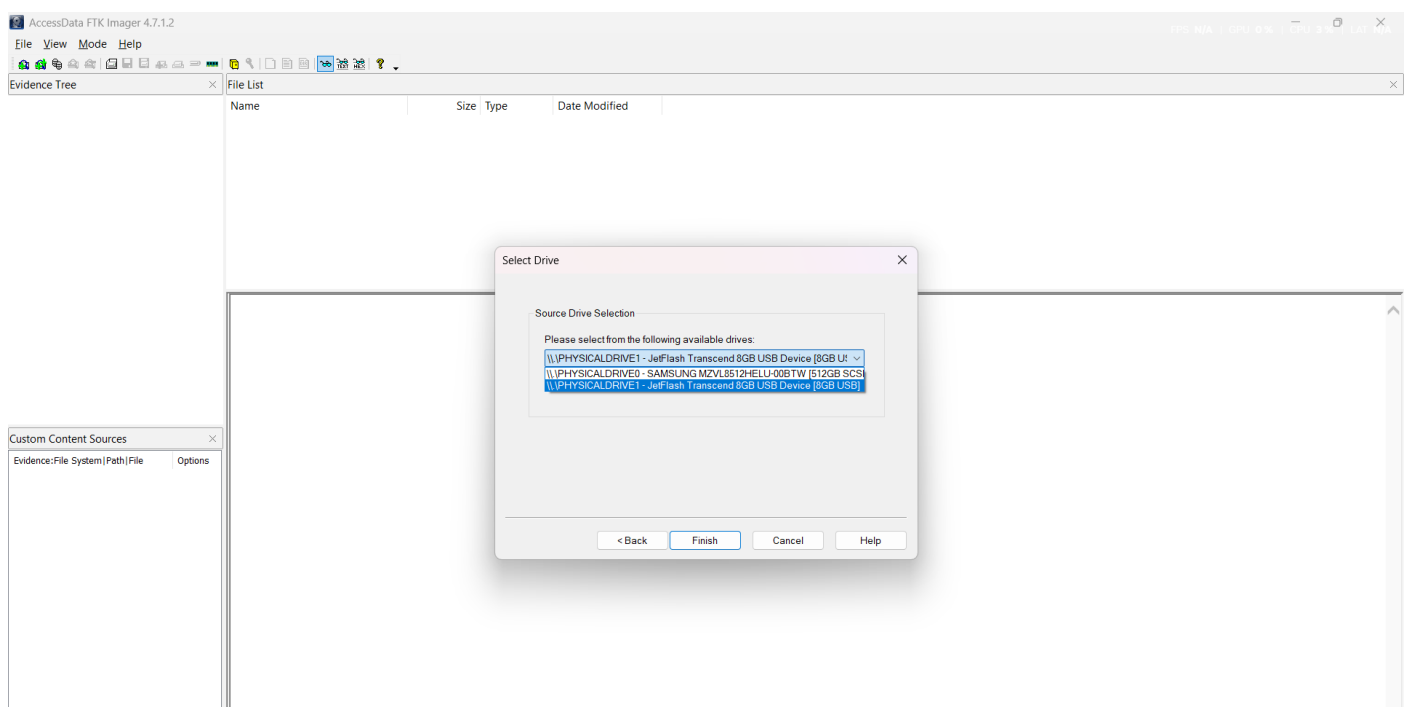


Figure 9: Selecting the JetFlash Transcend 8GB USB evidence drive.

Step 3: Select the Image Format

The destination image format was set to:

text

Raw (dd)

Raw/dd creates a bitstream image that can be processed by forensic-analysis tools.

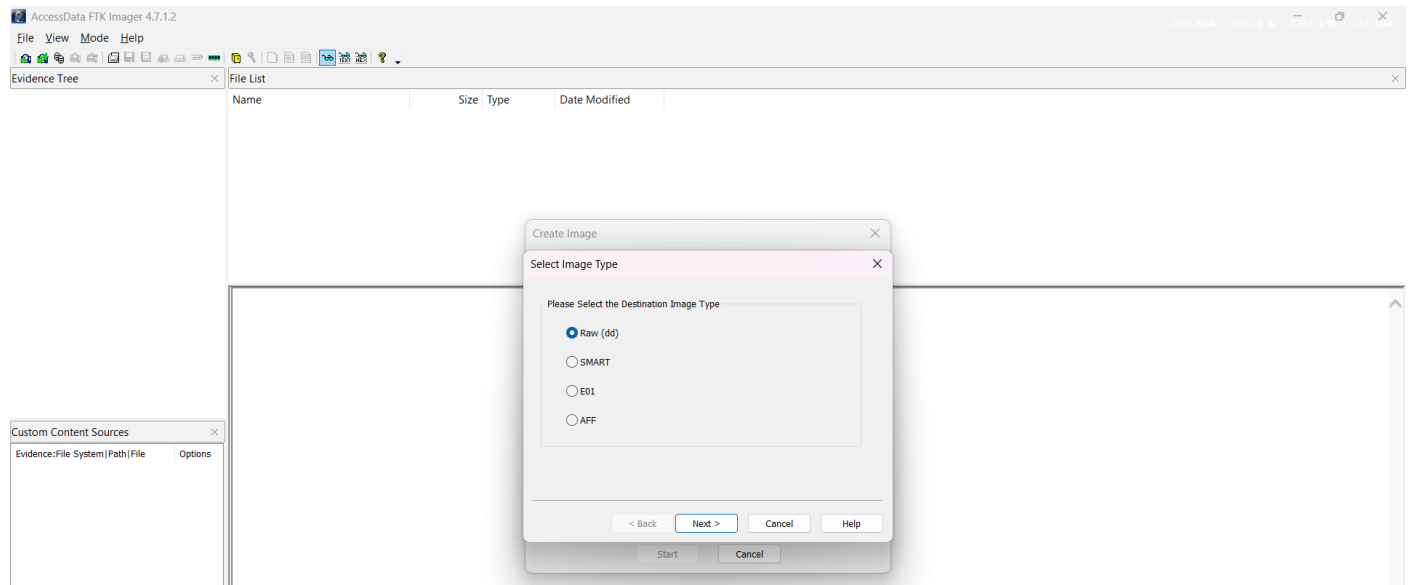


Figure 10: Selecting Raw (dd) as the destination image format.

Step 4: Enter Evidence Information

The evidence information shown in the FTK Imager screenshots was:

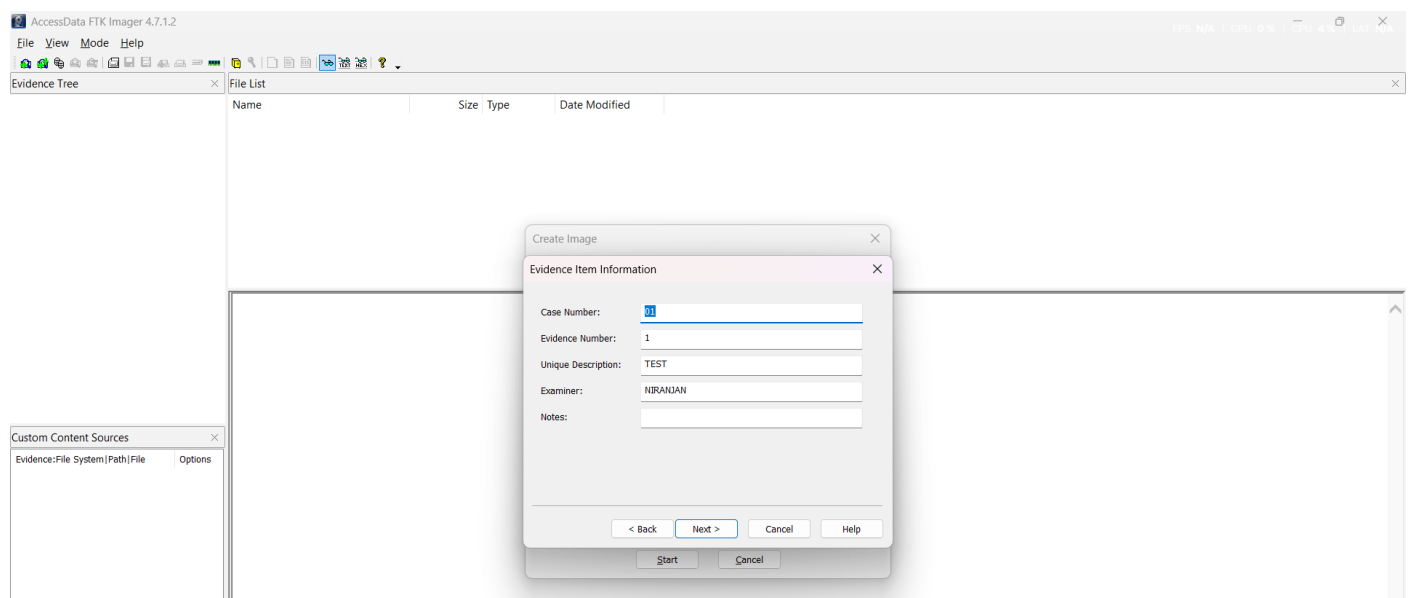


Figure 11: Entering case and evidence information.

Step 5: Acquire the Physical Drive

The image was created at:

text

D:\n\df\OP

The source device contained approximately 7683 MB of source data and had 15,734,784 sectors, with 512 bytes per sector.

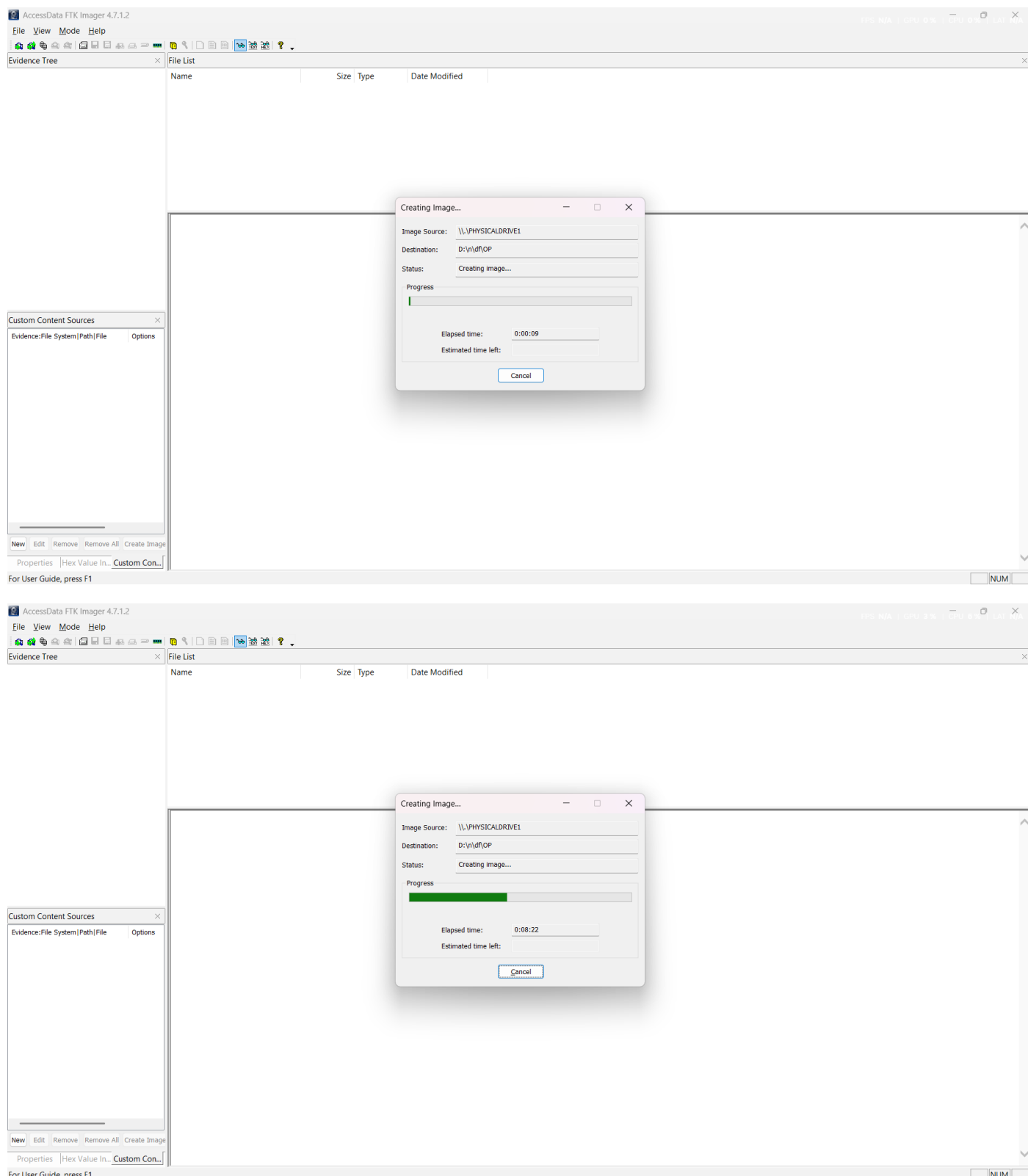


Figure 12: Physical-drive image creation in progress.

The acquisition completed successfully after approximately 18 minutes. FTK Imager generated segmented Raw/dd image files beginning with:

text

D:\n\df\OP.001

D:\n\df\OP.002

D:\n\df\OP.003

D:\n\df\OP.004

D:\n\df\OP.005

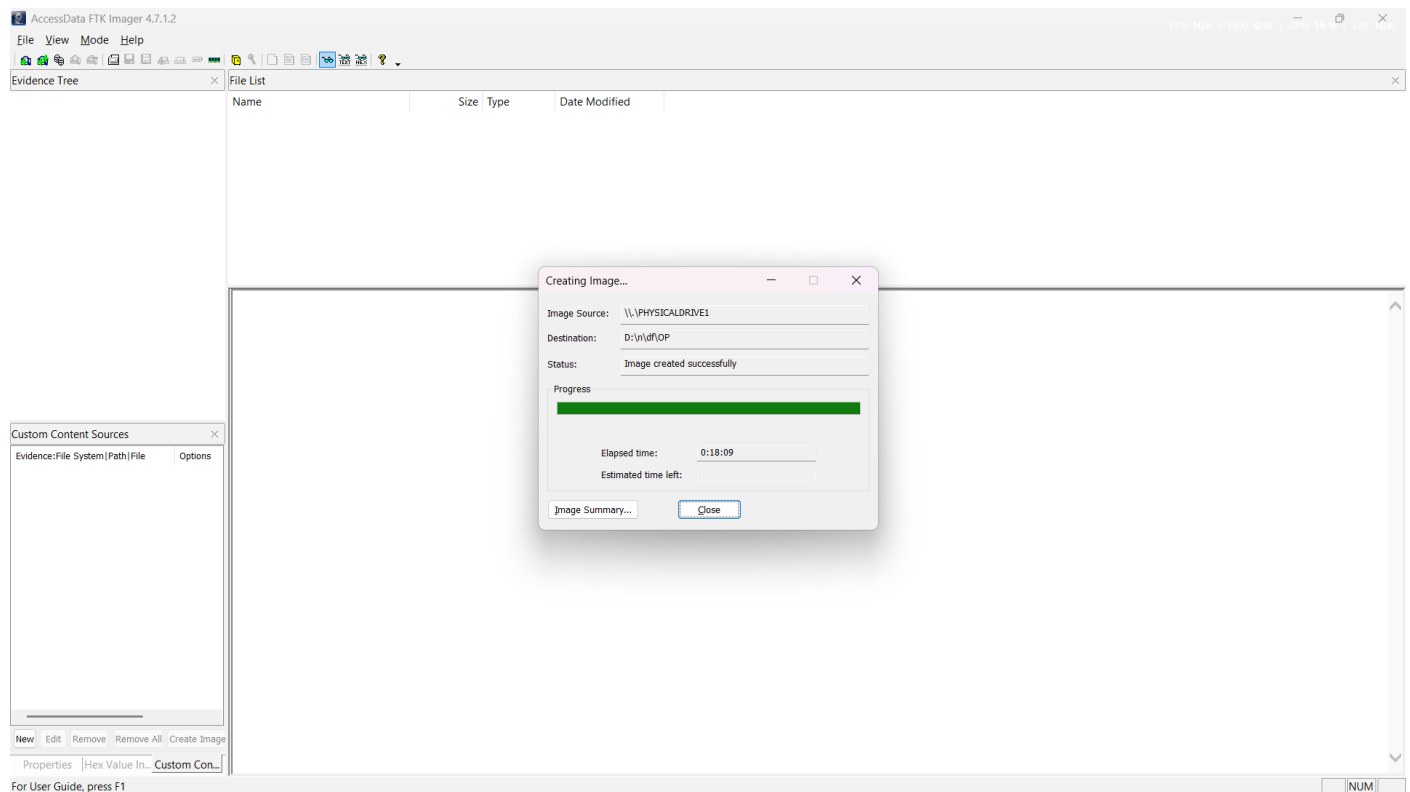


Figure 13: Successful completion of the physical-drive image creation.

6. Disk-Image Verification

FTK Imager calculated and compared the source and image hashes.

Hash	Value	Verification result
MD5	f199ed67d7bb3c51fde2c28b7b106ec4	Match
SHA1	9d90f654dab7ba6d35db7c1d571ce0165251a348	Match

The verification window also reported:

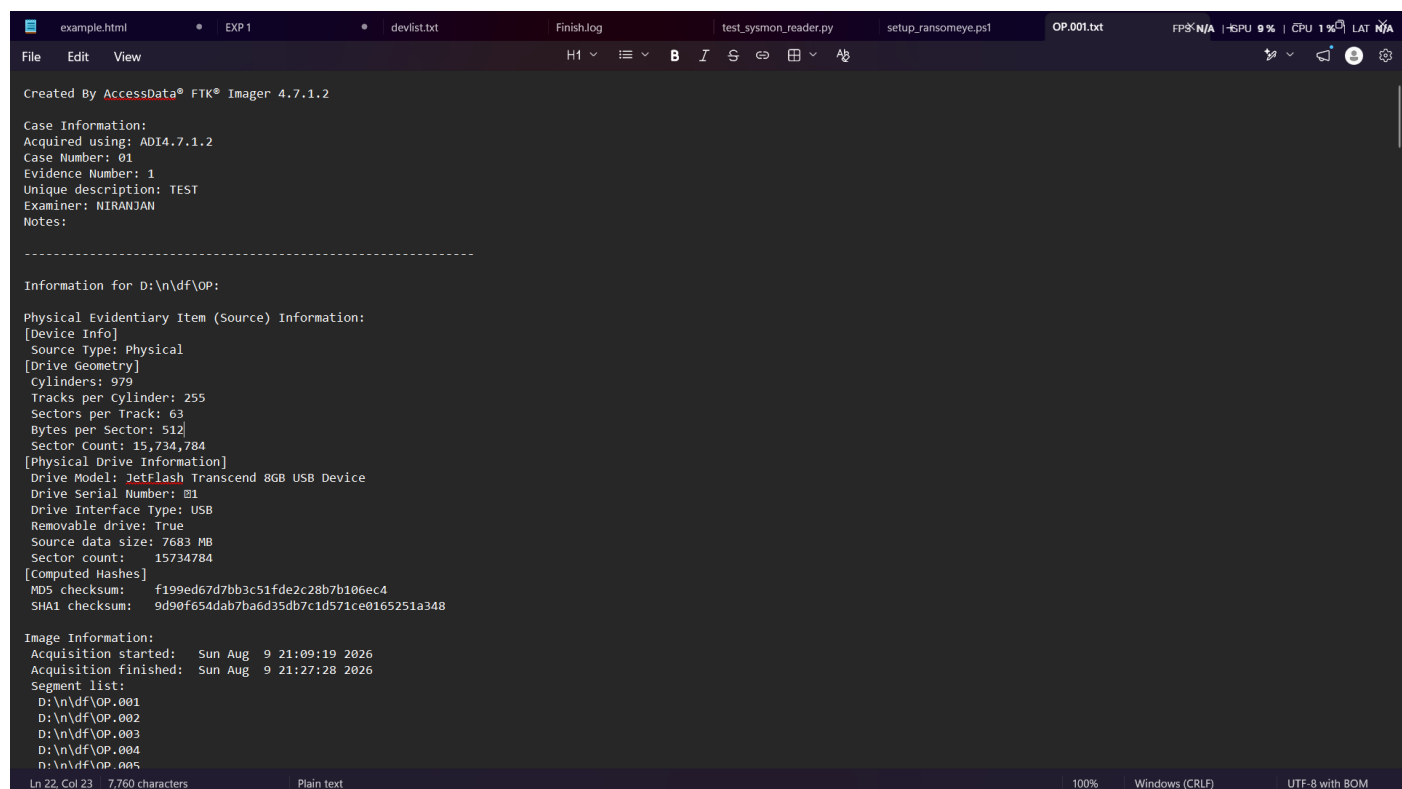
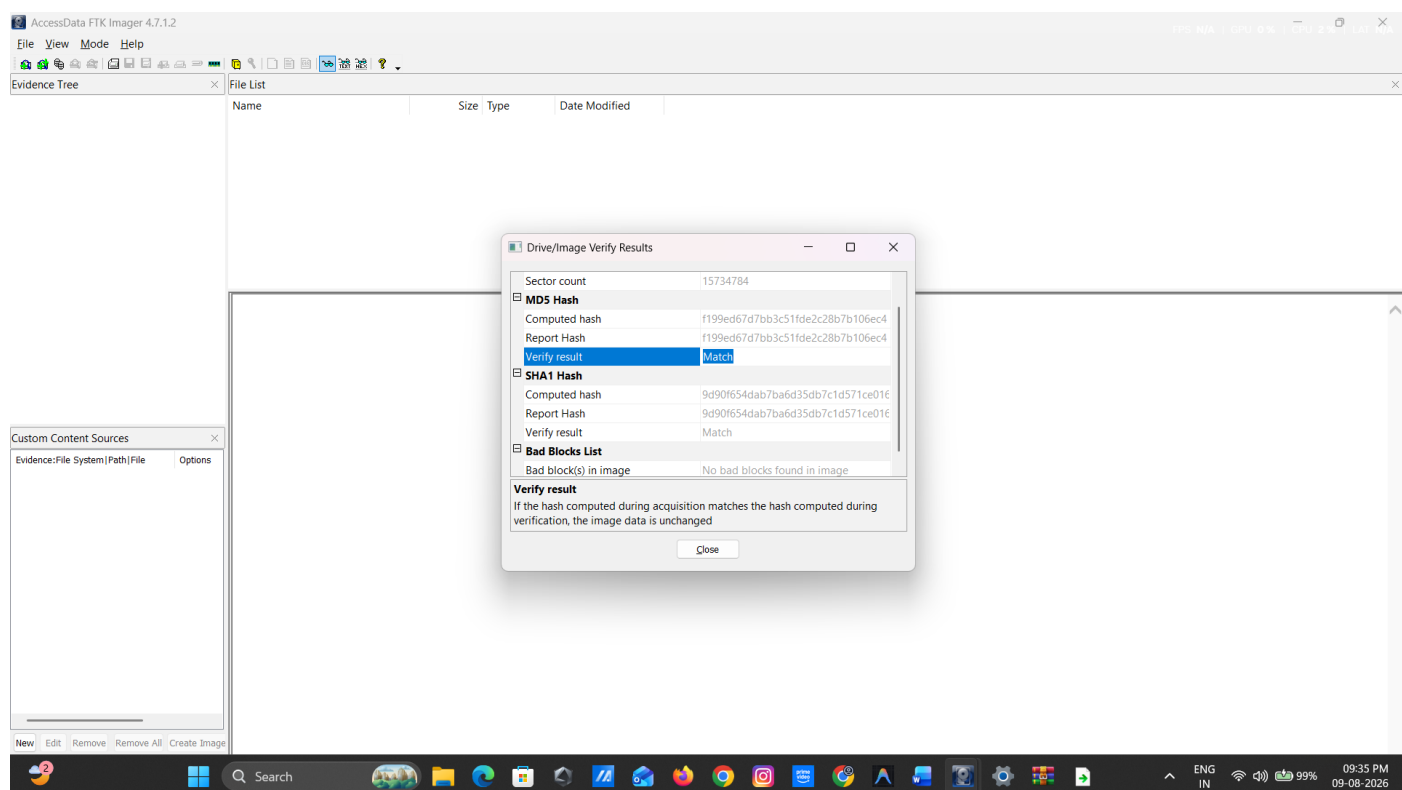


Figure 14: Drive/Image Verify Results showing MD5 and SHA1 Match.

7. Observations and Results

1. FTK Imager 4.7.1.2 was used successfully for both memory and physical-drive acquisition.
2. Volatile memory was saved as memdump.mem.
3. An AD1 evidence container was created as segmented files beginning with memcapture.ad1.
4. The pagefile option was enabled during memory capture.
5. The authorized JetFlash Transcend 8GB USB device was acquired from \\.\PHYSICALDRIVE1.
6. The physical drive was acquired using the Raw/dd format.
7. The disk-image output was saved in D:\n\df\OP as segmented files.
8. Disk-image MD5 verification returned Match.
9. Disk-image SHA1 verification returned Match.
10. No bad blocks were reported in the acquired image.
11. The memory-acquisition report recorded MD5 and SHA1 checksums for the captured memory evidence.

8. Result

Volatile memory and non-volatile storage evidence were successfully acquired using AccessData FTK Imager. The RAM capture produced the memdump.mem file and the segmented memcapture.ad1 evidence container. The JetFlash Transcend 8GB USB device was acquired as a Raw/dd disk image.

The disk-image verification produced matching MD5 and SHA1 values, and no bad blocks were found. This confirms that the acquired disk image is consistent with the source evidence.

9. Conclusion

The complete FTK Imager evidence-acquisition procedure was successfully performed. Volatile memory was captured from the live Windows system, and a physical image of the authorized USB evidence device was created. FTK Imager generated acquisition reports and cryptographic hashes for evidence documentation. The matching MD5 and SHA1 values confirm the integrity of the non-volatile forensic image.